EcoRoute: Analysis & Build Reference

> **Source:** *EcoRoute: AI Smart Waste Management & Dynamic Collection Routing for Subdivisions* — Capstone Research Proposal, University of Cebu – Banilad (November 2025)
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1. Executive Summary

EcoRoute is an **AIoT-based smart waste management system** designed specifically for **private residential subdivisions and HOAs** in the Philippines. It combines IoT smart-bin sensors, AI-powered fill-level prediction (LSTM), and dynamic route optimization (CVRP via Google OR-Tools) to replace inefficient fixed-schedule waste collection with a **proactive, data-driven, collect-when-needed** model.

The system aims to reduce overflow incidents by 70%, cut fuel consumption and travel distance by 20–33%, and provide real-time visibility into waste operations for administrators, haulers, and residents.

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2. Problem Statement

Problem	Evidence
Global waste rising to **3.4B tonnes by 2050**	World Bank, 2025
Philippines generates **40,000+ tons/day** of solid waste	DENR, 2024
Trucks run at **<60% capacity** on fixed routes	Gomez & Associates, 2023
Less than **75% of waste** properly collected in major PH cities	NSWMC, 2024
Cebu City spends est. **₱50M/year** on overflow-related costs	Lim, 2024
33%+ of global MSW is mismanaged due to insufficient collection	UNEP, 2025

The core issue is a **logistical mismatch**: trucks collect nearly-empty bins while others overflow, wasting fuel, labor, and creating unsanitary conditions. Fixed-schedule collection cannot adapt to variable waste generation rates across a subdivision.

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3. Research Objectives

General Objective: Design and evaluate an AI-enabled, IoT-based smart waste collection system that detects when subdivision waste sites need service and generates dynamic collection routes.

Specific Objectives:

1. Gather baseline data on waste generation and operational inefficiencies within target subdivisions
2. Develop IoT smart-bin hardware with a **hybrid sensing module** (Ultrasonic + Time-of-Flight for error validation) on an ESP32 MCU with cellular connectivity
3. Build an **AI module** that predicts fill-level trajectories and time-to-threshold per bin, with automated alerting for impending overflows and sensor anomalies
4. Implement a **maintenance dashboard** and **mobile interface** for the dynamic routing engine that optimizes collection schedules by targeting only bins predicted to breach thresholds

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4. Key Differentiators (vs. Bigbelly, Sensoneo, WasteHero, Routeware)

Feature	EcoRoute	Competitors
Predictive Fill Levels (ML/LSTM)	✅	❌ (all reactive)
HOA / Subdivision Focused	✅	❌ (all city/municipal)
Mobile App for Residents	✅	❌
Low-Cost Retrofit (existing bins)	✅	❌ (Bigbelly ~$5K/unit)
Solar-Powered Hardware	✅	Only Bigbelly
Integrated Billing / PoS	✅	❌
Cellular Independence (no Wi-Fi needed)	✅	Mixed
Scalable SaaS Model	✅	Mixed

Core Innovation: EcoRoute is the only system that integrates **predictive intelligence** (LSTM time-to-threshold), **IoT hardware**, **dynamic CVRP routing**, and an **HOA-centric ecosystem** into a single SaaS platform.

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5. System Architecture

5.1 High-Level Flow

Smart Bin Sensors → Cellular/MQTT → Cloud Ingestion → AI Prediction → Route Optimization → Dispatch → Driver PWA → Proof of Service → Analytics

5.2 Three-Layer Architecture

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┌──────────────────────────────────────────────────────────┐
│ FIELD LAYER (IoT Nodes)                                  │
│ ESP32 + Ultrasonic/ToF Sensor + GSM/LTE Module           │
│ Solar Panel + Battery Pack                               │
│ Firmware: Sampling → Filtering → MQTT Publish            │
│ Deep Sleep + OTA Updates                                 │
└──────────────────────────┬───────────────────────────────┘
                           │ MQTT over TLS (Cellular)
                           ▼
┌──────────────────────────────────────────────────────────┐
│ CLOUD LAYER                                              │
│ MQTT Broker (EMQX/Mosquitto)                             │
│ Ingestion Service → Time-Series DB + PostGIS             │
│ AI Service (LSTM Forecasting)                            │
│ Routing Service (Google OR-Tools + Maps API)             │
│ LLM+RAG Insights Pipeline                                │
│ REST API Gateway (HTTPS + RBAC)                          │
└──────────────────────────┬───────────────────────────────┘
                           │ HTTPS
                           ▼
┌──────────────────────────────────────────────────────────┐
│ APPLICATION LAYER                                        │
│ Admin Web Dashboard (ReactJS)                            │
│ Crew Mobile PWA (React Native / Expo)                    │
│ Resident App (React Native)                              │
└──────────────────────────────────────────────────────────┘
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5.3 Network Topology

Star topology with the cloud platform as the central hub. All IoT devices, admin dashboards, and mobile apps connect independently via cellular or Wi-Fi to the cloud. No complex on-site networking hardware required.

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6. Main Functionality & Business Process Workflow

6.1 End-to-End Waste Collection Workflow

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1. SENSE       → Smart bin sensors read fill level every N minutes
2. TRANSMIT    → ESP32 publishes telemetry via MQTT over cellular
3. INGEST      → Cloud validates, stores in time-series DB
4. PREDICT     → LSTM model forecasts time-to-threshold per bin
5. ALERT       → System flags bins approaching threshold (e.g., 80%)
6. OPTIMIZE    → Routing engine generates CVRP-optimized routes
7. DISPATCH    → Admin reviews/approves, assigns route to crew
8. NAVIGATE    → Driver follows turn-by-turn directions on PWA
9. COLLECT     → Driver arrives, empties bin
10. VERIFY     → Driver submits Proof of Service (NFC/photo/geo-tag)
11. REPORT     → KPI dashboards update; AI generates insights
12. REPEAT     → Continuous monitoring loop
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6.2 Administrator Modules (Web Dashboard)

Module	Functions
Account Management	Create, update, disable/archive user accounts; RBAC (admin, dispatcher, driver)
Bin & Device Management	Register new smart bins, view real-time status & health, update bin details/location, archive/remove bins
Route Planning & Dispatch	Generate AI-optimized routes (OR-Tools CVRP), assign routes to maintenance crews, monitor route progress in real-time
Notification Management	View system alerts (overflow, low battery, sensor anomaly, offline), configure alert thresholds per bin/zone
Reports & Analytics	KPI dashboard (distance reduction, overflow incidents, route adherence, sensor uptime), historical trends, export reports, AI Insights panel (LLM+RAG: hotspots, peak days, staffing tips)
System Configuration	Set depots, zones, device defaults, quiet hours, threshold percentages

6.3 Maintenance Crew Modules (Mobile PWA)

Module	Functions
Route Execution	View assigned route with stop sequence, navigate to each bin (turn-by-turn via Google Maps), mark bin as collected
Proof of Service (PoS)	Scan bin NFC tag or capture geo-tagged photo, add notes, upload for verification
Issue Logging	Report damaged bins, faulty sensors, vehicle breakdowns, access issues (locked gates) — triggers admin alerts

6.4 User Roles & Access

Role	Platform	Permissions
Admin	Web Dashboard	Full system access: CRUD users/bins, generate routes, view analytics, configure system
Dispatcher	Web Dashboard	Generate & assign routes, monitor progress, manage alerts
Driver	Mobile PWA	View assigned routes, navigate, submit PoS, report issues

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7. Technology Stack (Build Instructions)

7.1 Stack Overview

Layer	Technology	Purpose
Mobile	React Native + TypeScript + Expo	Cross-platform crew & resident apps
Web	ReactJS + TypeScript + Vite + Tailwind CSS + shadcn/ui	Admin dashboard
Backend API	TypeScript / Bun + HonoJS	Primary REST API server
AI/ML Service	Python + FastAPI + Gunicorn	LSTM forecasting, AI insights
Route Optimization	Google OR-Tools + Celery + Redis	CVRP dynamic route solver

Layer	Technology	Purpose
IoT Firmware	C/C++ (Arduino IDE) + ESP32	Sensor reading, MQTT publish, power mgmt
MQTT Broker	EMQX or Mosquitto	IoT message transport
Database	Supabase (PostgreSQL + PostGIS)	Relational + geospatial + time-series storage
Maps & Routing	Google Maps API (Distance Matrix + Routes)	Distance/ETA calculations, turn-by-turn nav
AI Insights	LLM + RAG pipeline	Natural language operational recommendations
HTTP Client	Axios or Fetch API	Frontend-backend communication
Infrastructure	VPS (DigitalOcean/Hetzner) + Docker Compose	Container orchestration & deployment
Auth	Supabase Auth (or Firebase Auth)	Authentication + RBAC

7.2 IoT Hardware BOM

Component	Specification	Purpose
Microcontroller	ESP32 dev board	Main controller: sensing, MQTT, power management
Primary Sensor	HC-SR04 / JSN-SR04T (Ultrasonic)	Fill-level distance measurement
Validation Sensor	VL53L0X (Time-of-Flight laser)	Cross-validates ultrasonic; eliminates false readings
Connectivity	GSM/LTE module + SIM	Cellular MQTT transport (no Wi-Fi dependency)
Power	Solar panel + Li-Ion battery pack (or DC adapter)	Self-powered outdoor operation
Enclosure	IP-rated outdoor box	Weather protection
Extras	Status LED, optional buzzer, voltage regulator	Feedback & power regulation
Programming	Arduino IDE + USB-to-serial	Firmware development & flashing

7.3 Mobile Hardware Requirements

Component	Minimum
CPU	Octa-core or equivalent
RAM	3 GB+
Storage	32 GB+
OS	Android 10+ / iOS 14+
Display	5"+ at 720p
Connectivity	4G LTE, Wi-Fi 4+, GPS
Sensors	Camera (for PoS photo capture)

7.4 Development Tooling

Tool	Use
VS Code	Primary IDE for web/backend
Android Studio	Android builds & testing
Xcode	iOS builds & testing
Arduino IDE	ESP32 firmware development
Docker / Docker Compose	Service containerization
Git	Version control

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8. Database Schema (PostgreSQL + PostGIS via Supabase)

Normalized to **3NF**. Key entities:

Entity	Purpose	Key Fields
subdivision	Multi-tenant root (HOA/community)	subdivision_id, name, code, geofence (Polygon)
user	Auth + RBAC	user_id, subdivision_id (FK), email, role (admin/dispatcher/driver)
smart_bin	Digital twin of IoT device	device_id, device_code (IMEI), location_point (PostGIS Point), capacity_liters, threshold_percent, status
bin_telemetry	High-volume time-series sensor data	device_id (FK), fill_level_percent, distance_cm, battery_voltage, signal_strength, anomaly_flag
fill_prediction	AI forecast outputs	device_id (FK), predicted_fill_percent, time_to_threshold_minutes, confidence_score, model_version
alert	System events needing attention	alert_type (overflow/low_battery/sensor_anomaly/offline), severity, is_acknowledged
collection_route	Daily job order / dispatch manifest	status (planned/in_progress/completed), optimization_score, estimated_distance_km, assigned_driver_id, assigned_vehicle_id
route_stop	Junction: Route ↔ Bin (M:N)	route_id (FK), device_id (FK), sequence_order, status (pending/arrived/serviced/skipped), photo_proof_url
service_event	Proof of Service execution	device_id (FK), driver_id (FK), event_type, location (PostGIS Point), photo_urls (JSON), service_duration_sec
audit_log	Immutable action trail	user_id, action, entity_type, old_values (JSON), new_values (JSON), ip_address
notification	Outbound message queue	user_id (FK), channel (push/sms/email/in_app), title, body, is_read
system_config	Key-value settings store	subdivision_id (FK, nullable for global), config_key, config_value

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9. AI / ML Components

9.1 Fill-Level Forecasting

- **Model:** LSTM (Long Short-Term Memory) neural network

- **Input:** Historical time-series fill-level data from `bin_telemetry`
- **Output:** `predicted_fill_percent` and `time_to_threshold_minutes` stored in `fill_prediction`
- **Purpose:** Proactive scheduling — predict when a bin will reach 80% capacity *before* it overflows
- **Alternatives considered:** ARIMA (classical), 1D CNN, GRU, Bi-LSTM — literature (Kolekar et al., 2023) favors LSTM for resilience and accuracy

9.2 Dynamic Route Optimization

- **Framework:** Google OR-Tools (CVRP / VRPTW solver)
- **Input:** List of bins predicted to breach threshold + truck capacity + depot location + Google Maps distance matrix
- **Output:** Optimized route sequence minimizing total distance/time
- **Constraint model:** Capacitated Vehicle Routing Problem (vehicles have fixed capacity)

9.3 AI Insights (LLM + RAG)

- **Purpose:** Translate complex telemetry patterns into natural language recommendations
- **Examples:** Hotspot identification, peak day analysis, staffing suggestions, ESG/KPI reports
- **Architecture:** RAG pipeline over historical telemetry and route data

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10. Module Ownership & Sprint Assignment

Developer	Modules (24 total)	Points
Roel	Account Management (4), Bin & Device Management (4)	8
Jhonn	Route & Dispatch Management (3), Route Execution (3), Proof of Service & Issues (2)	8
Sebastian	Notification Management (2), Reports & Analytics (4), AI Insights (1)	8*

*Points adjusted for balance; paper lists 7+1 = 8 for Sebastian across admin+maintenance modules.

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11. Methodology

Agile SDLC with 6 iterative phases per sprint:

1. **Requirements** — Stakeholder interviews, site surveys, baseline data gathering

2. **Design** — Architecture diagrams, ERD, wireframes, UI prototypes
3. **Development** — Firmware, backend services, frontend apps (CI/CD, code reviews)
4. **Testing** — Unit, integration, field tests; sensor accuracy, connectivity, route correctness, UAT
5. **Deployment** — Pilot rollout (10–15 bins), SIM provisioning, HTTPS/RBAC enforcement, training
6. **Review** — Retrospective, KPI assessment, threshold tuning, iteration planning

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12. Success Criteria

KPI	Target
Overflow incident reduction	≥ 70%
Fuel use / travel distance reduction	≥ 20%
Route adherence	Tracked via GPS
Sensor uptime	Monitored continuously
Resident / HOA satisfaction	Positive pilot feedback

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13. Scope & Limitations

In Scope: Residential subdivisions, outdoor smart bin lids, hybrid ultrasonic/ToF sensing, ESP32 + cellular MQTT, LSTM forecasting, CVRP routing, admin dashboard, crew PWA, PoS.

Limitations:

- ML model trained on limited sites — transferability may require retraining
- Garbage condition (compacted, large items) can obstruct sensor accuracy
- System depends on cellular signal quality — prolonged outages delay data
- Pilot-scale deployment; city-level scaling requires further validation

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14. Business Model

Revenue	Cost
Per-bin/month subscription (monitoring, routing, dashboards)	Hardware BOM (ESP32, sensors, enclosures, batteries)
One-time setup/installation fee per site	Assembly/installation labor
Analytics add-on (LLM insights, ESG/KPI)	Cellular SIM connectivity
Maintenance contracts	Cloud hosting/storage
Hardware sale or lease (sensor kits)	Google Maps API usage
White-label licensing for contractors/LGUs	R&D, QA, support

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15. Quick-Start Build Checklist

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[ ] Set up monorepo (Turborepo or Nx)
[ ] Initialize Supabase project (PostgreSQL + PostGIS + Auth)
[ ] Scaffold backend API (Bun + HonoJS)
[ ] Scaffold Python ML service (FastAPI + Gunicorn)
[ ] Set up MQTT broker (EMQX or Mosquitto via Docker)
[ ] Set up Redis + Celery for async route optimization
[ ] Scaffold admin dashboard (Vite + React + TypeScript + Tailwind + shadcn/ui)
[ ] Scaffold mobile app (React Native + Expo + TypeScript)
[ ] Create database migrations for all 12 entities
[ ] Configure Docker Compose for all services
[ ] Set up Google Maps API (Distance Matrix + Routes)
[ ] Set up Google OR-Tools integration
[ ] Flash ESP32 firmware (Arduino IDE) with ultrasonic/ToF + MQTT
[ ] Implement RBAC middleware (admin/dispatcher/driver)
[ ] Build telemetry ingestion pipeline (MQTT → validate → store)
[ ] Train LSTM model on simulated/collected fill-level data
[ ] Build route generation endpoint (OR-Tools CVRP solver)
[ ] Build PoS upload flow (photo + geo-tag + NFC)
[ ] Build notification system (push/SMS/email/in-app)
[ ] Build analytics dashboard with LLM+RAG insights
[ ] Deploy to VPS (DigitalOcean/Hetzner) with HTTPS
[ ] Pilot test with 10–15 bins in target subdivision
```